Pankaj Bhattacharjee

9564287282 | pankajbhattacharjee634@gmail.com | LinkedIn | GitHub | Portfolio | Kolkata, India

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering 2023-2027
Brainware University, Kolkata | CGPA: 7.00/10

Internship Experience

Artificial Intelligence Intern | Infosys Springboard | Remote Sep – Nov 2025

- Built an **AI content marketing system using NLP, Python, and Machine Learning models**, achieving **86%** accuracy.
- Implemented sentiment analysis and **A/B testing** to evaluate content performance, improving engagement by **24%**.
- Tracked campaign metrics using **Google Sheets** and improved model performance through **periodic retraining**. [Certificate]

Machine Learning Intern | Unified Mentor Pvt. Ltd | Remote Apr – May 2025

- Developed a fraud detection model using **Random Forest** on financial transaction data, achieving **91% classification accuracy**.
- Performed data preprocessing, model training, and evaluation using confusion **matrix** and **ROC-AUC** metrics.
- Built a **Streamlit** web application for real-time fraud detection demonstration. [Certificate]

Artificial Intelligence Intern | Samsung Innovation Campus Aug – Oct 2025

- Developed **demand forecasting models** using time series and ensemble techniques on large datasets (100K+ records).
- Improved prediction accuracy by **28%** through feature engineering and model tuning.
- Integrated trained models into **Python-based data pipelines** for deployment and testing. [Certificate]

Technical Skills

Languages: Python, SQL, C, PostgreSQL, MongoDB, Redis
Machine Learning & AI: scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, NLP (spaCy, NLTK)

Generative AI & Agentic AI: LLMs, Prompt Engineering, RAG, AI Agents, LangChain, LangGraph, FAISS, ChromaDB, Conversational AI.

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Deep Learning: CNN, RNN, LSTM, Transformers (HuggingFace), Generative AI (LLM APIs)

Backend & Deployment: Flask, FastAPI, REST APIs, Docker, Basic Cloud (AWS, GCP)

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Kaggle, Streamlit

Projects

RAG-Powered Math Problem Solver & Tutor | [Live Demo] | [GitHub] Feb – Mar 2026

Python, LangChain, LangGraph, FAISS, Gemini API, FastAPI, Streamlit, HuggingFace Transformers

- Built an **end-to-end Retrieval-Augmented Generation (RAG) pipeline** that answers **multi-step math** problems (algebra, calculus, statistics) by retrieving relevant theorems and formulas from a curated **vector knowledge base using FAISS**.
- Designed an **agentic reasoning workflow** with **LangGraph** that breaks complex problems into sub-steps, solves each using an **LLM**, and synthesizes a final explained solution — achieving **89% correctness on a 500-question benchmark**.
- Integrated **HuggingFace sentence-transformers** for **semantic embedding** of math content and built a **FastAPI backend** with a **Streamlit frontend** for **real-time interactive tutoring** with step-by-step explanations.
- Implemented **conversation memory** using **ChromaDB** to maintain **multi-turn dialogue context**, enabling the assistant to reference earlier parts of a tutoring session for coherent follow-up answers.

Certifications and Achievements

- **GenAI Exchange Hackathon – Hack2skill (2025):** Designed an **LLM-powered marketplace assistant** and **shortlisted in the initial evaluation round**. [Link]
- **Internshala Student Partner (ISP) – Internshala (2026):** Selected as a **campus representative** to promote internships and training programs.
- **Infosys Certifications (2025):** Completed certifications in Computer Vision**,** Natural Language Processing**,** AI/ML**, and** Deep Learning.
- **AWS Solutions Architecture Simulation – Forage (2025)** [Link]
- **Deloitte Data Analytics Simulation – Forage (2025)** [Link]
- **TATA Data Analytics Simulation – Forage (2025)** [Link]
- **Nestlé e-Learning – Python & Machine Learning (2025)** [Link]